Establishing Genomics Infrastructure for Australian Researchers

Tiffanie M. Nelson,[1*] Jeffrey H. Christiansen,[1] Ziad Al Bkhetan,[1] Catherine Bromhead,[1] Melissa Burke,[1] Patrick Capon,[1] Keeva Connolly,[1] O. Johan R. Gustafsson,[1] Mark Gray,[2] Dominique Gorse,[3] Christina Hall,[1] Kathryn Hall,[4] Cameron Hyde,[3] Farah Z. Khan,[1] Justin Lee,[2] Steven Manos,[1] Christopher Mangion,[3] Igor Makunin,[3] Winnie Mok,[1] Lisa Phippard,[1] Gareth Price,[1] Anna Syme,[1] Mike W. C. Thang,[3] Nigel Ward,[1] and Andrew Lonie[1]

[1]Australian BioCommons, Australia. [2]Pawsey Supercomputing Research Centre, Perth, Australia.
[3]Queensland Cyber Infrastructure Foundation. [4]Atlas of Living Australia, CSIRO
* tiff@biocommons.org.au

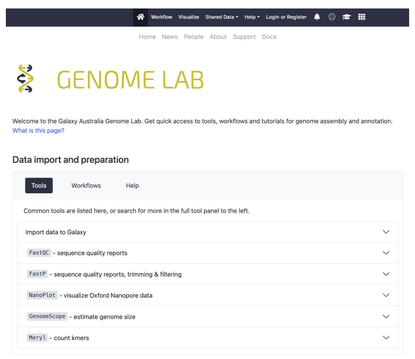
Since 2020, Australian BioCommons has developed community-scale digital capacity, training and bioinformatics infrastructure to support Australia's life scientists. By engaging with the national community of practitioners, we have distilled their community-level roadblocks and needs, and have established numerous computational systems and services to address them.

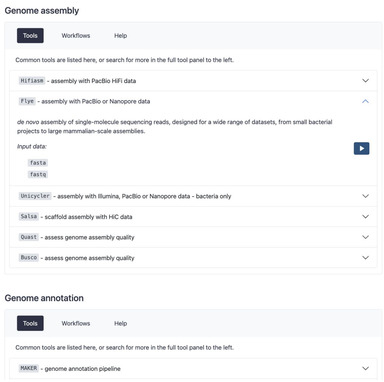
Over **8 million jobs** run and **27,000+ users** on Galaxy Australia

FGENESH++ and Apollo Services cater to **100+ active users**

Galaxy Australia's Genome Lab

- A new user-friendly place to use Galaxy Australia customised for individuals undertaking genomics research providing quick and easy access to the most relevant tools, workflows and help, **genome.usegalaxy.org.au**





Automatic and Manual Genome Annotation Services

- An automatic genome annotation software pipeline which predicts genes in eukaryotic genomes
- Choose from **500+** eukaryote matrices
- Run on Galaxy Australia or via the command line, **biocommons.org.au/Fgenesh-Plus-Plus**

- Manually curate genomes collaboratively or privately in a hosted Apollo server for your lab/consortium built and maintained for you
- Draft genomes can be privately shared or made public via JBrowse
- Access via application at: **apollo-portal.genome.edu.au/**

Support with Compute and Workflows

- Computational resources to generate valuable data assets via the Australian BioCommons Leadership Share (ABLeS), **biocommons.org.au/ables**
- Discover **71 optimised and reusable workflows** on the BioCommons WorkflowHub space, **workflowhub.eu/programmes/8**
- Join **32 research groups** on the Nextflow Seqera Service to run and monitor the execution of Nextflow pipelines on your compute infrastructure, **biocommons.org.au/seqera-platform**
- How-to Guides, **australianbiocommons.github.io/how-to-guides**. and workflows to run on Galaxy Australia, including the Vertebrate Genomes Project assembly workflow, **https://github.com/AnnaSyme/wfs-VGP**

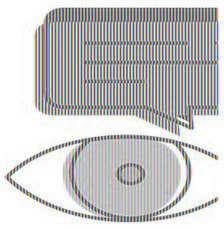
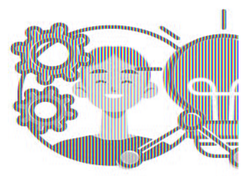
Support for Tool and Data Discovery and Data Deposition

- A one stop shop for genomic data of species relevant to Australia, **arga.org.au**
- Discover indexed genomic data relevant to Australian taxa
- Easier data uploads to international repositories via the ENA Upload tool, **github.com/usegalaxy-eu/ena-upload-cli**
- Find **100's of tools** and containerised software easily via ToolFinder, **biocommons.org.au/tool-finder**

Sign up to our monthly newsletter, **biocommons.org.au/subscribe** 👉

Skills, Training and Engagement

- Webinars, and workshops delivered routinely **29 workshops** to **3,700+ people** since 2019, **biocommons.org.au/webinars-workshops**
- More than **105k views** on YouTube **youtube.com/AustralianBioCommonsChannel**



- Training materials shared in Zenodo **zenodo.org/communities/australianbiocommons-training**
- Join a research-domain focused mailing groups, **biocommons.org.au/get-involved**